

Yanzhou Fu

Mechanical Engineering Ph.D.

Florida Atlantic University
777 Glades Road Boca Raton, FL 33431

✉ fuy@fau.edu

🌐 <https://github.com/VISTA-Laboratory>

🌐 www.linkedin.com/in/yanzhou-fu-949b3b1a3/

Education

- 2019 - 2023 **Ph.D.**, Mechanical Engineering, University of South Carolina, Columbia, SC.
• **Additive Manufacturing** • **Artificial Intelligence** • **Structural Validation**
- 2012 - 2015 **M.S.**, Mechanical Engineering, Northeast Agricultural University, China.
- 2008 - 2012 **B.S.**, Mechanical Engineering, Heze University, China.

Work Experience

- 2026.01 - Present **Assistant Professor**, Departments of Ocean and Mechanical Engineering, Florida Atlantic University, Boca Raton, FL.
- 2025.01 - 2026.12 **Research Assistant Professor**, Departments of Mechanical Engineering, University of South Carolina, Columbia, SC.
- 2024.10 - 2025.12 **NSF eFellows funded Postdoctoral Fellow**, advisor: Gurcan Comert, Computer Science, Physics and Engineering Department, Benedict College, Columbia, SC.
- 2024.05 - 2024.10 **Research Associate**, Computer Science, Physics and Engineering Department, Benedict College, Columbia, SC.
- 2019.09 - 2023.12 **Graduate Research Assistant**, advisor: Austin Downey, Departments of Mechanical Engineering, University of South Carolina, Columbia, SC.
- 2015.08 - 2018.03 **Automation, Senior Engineer**, BOE Technology Group CO LTD, China.

Grants and Contracts

- [1] **National Science Foundation**, “EXCELLENCE in RESEARCH: Real-Time Monitoring and Structural Validation for Continuous Fiber Printing,” Aug 2025 – Jul 2028, \$887,740; **PI: Yanzhou Fu**, Co-PI: Austin Downey (USC), Co-PI: Simon Laflamme (ISU), Co-PI: Gurcan Comert (NCAT). (Funded)
- [2] **Department of Transportation (DOT) – Federal Motor Carrier Safety Administration (FMCSA)**, “Real-Time Topology-Aware Artificial Intelligence for Tire Health Monitoring Systems with Soft Sensor Array,” Oct 2025 – Sep 2027, \$1,022,614; **PI: Gurcan Comert**; **Co-PI: Yanzhou Fu**. (Funded)
- [3] **National Aeronautics and Space Administration**, “Event Camera paired with Time-series AI to Enable In-Space Melt Pool Monitoring for Metal-based Additive Manufacturing,” May 2025 – May 2026, \$20,000; **PI: Austin Downey (USC)**; **Co-PI: Yanzhou Fu**. (Funded)
- [4] **University of South Carolina**, “Development of a Low-Cost Wire Arc Additive Manufacturing System for Advanced Situ Monitoring Research and Education,” July 2025 – Nov 2026, \$15,000; **PI: Yanzhou Fu**. (Funded)
- [5] **University of South Carolina**, “Neuromorphic Spiking Neural Networks and Event Cameras for Edge-Based In-Situ Monitoring of Additive Manufacturing,” July 2025 – Nov 2026, \$30,000; **PI: Yanzhou Fu**. (Funded)
- [6] **National Aeronautics and Space Administration**, “Artificial Intelligence for Anomaly Detection in Laser Powder Bed Fusion Additive Manufacturing,” May 2024 – May 2025,

- \$35,000; PI: Austin Downey (USC); **Co-PI: Yanzhou Fu.** (Funded)
- [7] **National Science Foundation**, “Engineering Fellows Postdoctoral Fellowship Program,” Oct 2024 – Oct 2026, \$255,913; Advisor: Gurcan Comert (BC); **Fellow: Yanzhou Fu.** (Funded)
 - [8] **National Science Foundation**, “Travel Grants for Attending the ASME Rising Stars,” Nov 2024, \$940; **PI: Yanzhou Fu.** (Funded)
 - [9] **National Aeronautics and Space Administration**, “Advancing Gap Analysis and Data Assimilation for Air Quality Research through Classical and Quantum Machine Learning Comparisons,” May 2025 – May 2027, \$1,199,979; PI: Gurcan Comert (NCAT); **Co-PI: Yanzhou Fu**, Co-PI: Varghese Vaidyan (DSU). (Pending)
 - [10] **National Aeronautics and Space Administration**, “RFA-029, Adaptive Laser Welding Framework for Space Applications: Modeling, Monitoring, and RealTime Control in Vacuum and Microgravity Environments,” Sep 2025 – Aug 2026, \$125,000; PI: Lang Yuan (USC); **Co-PI: Yanzhou Fu.** (Pending)
 - [11] **Department Of Energy**, “Agentic AI Framework for Smart Welding: Bidirectional Digital Twin with System-Level Decision Support,” Jul 2026 – Mar 2027, \$750,000; **PI: Yanzhou Fu.** (Pending)

Research Projects

Additive Manufacturing In Situ Monitoring	In-situ monitoring systems provide real-time quality control in metal and plastic additive manufacturing. By tracking void defects and microstructural changes, this project supports high-stakes applications, ensuring structural reliability for aerospace and marine energy components.
Machine Learning for Defect Detection	Projects focus on enhancing defect detection in additive manufacturing using deep learning (CNNs) for precise image-based identification, SVMs for impact assessment, and U-Net for void segmentation. K-means clustering identifies anomalous splatter patterns, enabling proactive interventions essential for aerospace and ocean engineering applications.
Real-Time Sensing and Validation	Real-time validation frameworks with defect impact decision boundary and Abaqus Python scripting, enables rapid structural assessments. This approach, suited to extreme environments, ensures resilience and quality in additively manufactured components for aerospace and ocean engineering applications.
AI-Driven Decision-Making	AI-driven system autonomously assesses structural integrity by integrating finite element model data. The system enables real-time, automated decisions, improving quality control for additive manufacturing in mission-critical aerospace and marine applications.

Key Statistics

Grant Funding Requested	\$2.42 million in awarded and pending funding through grants from NSF, NASA, DOE, and the Department of Environment.
2019 - present	15 publications, 661 citations in machine learning-based structural validation for additive manufacturing. Three of them are published in Additive Manufacturing with Impact Factor 10.3.
210 citations	In situ monitoring for fused filament fabrication process: A review, cited in the Spring 2024 Army SBIR Program Solicitation , “Leveraging Advanced Computation to Better Employ Additive Manufacturing”.
373 citations	Machine learning algorithms for defect detection in metal laser-based additive manufacturing: A review

Journal Articles

- [1] **Yanzhou Fu**, Austin RJ Downey, Lang Yuan, Hung-Tien Huang, and Emmanuel A Ogunniyi. Simulation-in-the-loop additive manufacturing for real-time structural validation and digital twin development. *Additive Manufacturing (Impact Factor 10.3)*, page 104631, 2025
- [2] **Yanzhou Fu**, Austin RJ Downey, Lang Yuan, and Hung-Tien Huang. Real-time structural validation for material extrusion additive manufacturing. *Additive Manufacturing (Impact Factor 10.3)*, page 103409, 2023
- [3] **Yanzhou Fu**, Austin RJ Downey, Lang Yuan, Tianyu Zhang, Avery Pratt, and Yunusa Balogun. Machine learning algorithms for defect detection in metal laser-based additive manufacturing: A review. *Journal of Manufacturing Processes*, 75:693–710, 2022
- [4] **Yanzhou Fu**, Austin Downey, Lang Yuan, Avery Pratt, and Yunusa Balogun. In situ monitoring for fused filament fabrication process: A review. *Additive Manufacturing (Impact Factor 10.3)*, 38:101749, March 2021
- [5] Ali Zolfagharian, Liuchao Jin, Qi Ge, Wei-Hsin Liao, Andrés Díaz Lantada, Francisco Franco Martínez, Tianyu Zhang, Tao Liu, Charlie CL Wang, Mohammad Hossein Mosallanejad, et al. Roadmap on artificial intelligence-augmented additive manufacturing. *Advanced Intelligent Systems*, page e202500484, 2026
- [6] Joud N Satme, **Fu, Yanzhou**, Samuel Roberts, Austin RJ Downey, Tianyu Zhang, Lang Yuan, and Daniel Kiracofe. Tunable shock and vibration damping in metal laser powder bed fusion components via controlled post-print compaction of particle damper cavities. *Smart Materials and Structures*, 35(4):045015, 2026

Peer-Reviewed Conference Proceedings

- [1] Joud N Satme, **Yanzhou Fu**, Austin R Downey, Lang Yuan, and Daniel Kiracofe. Impact of particle packing density on the frequency response of an additively manufactured particle damper. In *AIAA SCITECH 2024 Forum*, page 0024, 2024

Conference Proceedings

- [1] **Yanzhou Fu**, Matt Whetham, Austin RJ Downey, Lang Yuan, and Gurcan Comert. A study of online melt pool, plume, and spatter tracking in laser powder bed fusion using dbscan. In *Digital Twins, AI, and NDE for Industry Applications and Energy Systems 2025*, volume 13438, pages 144–150. SPIE, 2025
- [2] Samuel Roberts, **Yanzhou Fu**, Joud N Satme, Daniel Kiracofe, Austin RJ Downey, and Lang Yuan. Modeling post-process indenting using the discrete element method for particle density control in additively manufactured dampers. In *Active and Passive Smart Structures and Integrated Systems XIX*, volume 13432, pages 255–261. SPIE, 2025
- [3] **Yanzhou Fu**, Joud N Satme, Austin R Downey, Lang Yuan, Tianyu Zhang, and Daniel Kiracofe. Investigating compressing particle damper pockets in beams manufactured by laser powder bed fusion additive manufacturing. In *IMAC-XLI*, volume 11592, pages 315–321. IMAC, 2023
- [4] **Yanzhou Fu**, Priddy Braden, Austin R Downey, and Lang Yuan. Real-time splatter tracking in laser powder bed fusion additive manufacturing. In *Nondestructive Characterization and Monitoring of Advanced Materials, Aerospace, Civil Infrastructure, and Transportation XV*, volume 192, pages 302–309. SPIE, 2023
- [5] **Yanzhou Fu**, Austin Downey, Lang Yuan, Avery Pratt, and Yunusa Balogun. In situ structural validation of components manufactured using fused filament fabrication. In *Nondestructive Characterization and Monitoring of Advanced Materials, Aerospace, Civil Infrastructure, and Transportation XV*, volume 11592, pages 315–321. SPIE, 2021

- [6] Digambar Killedar, Emmanuel Ogunniyi, Yanzhou Fu, Austin Downey, Lang Yuan, and Matthew D Folsom. In-situ monitoring of geometric drift and thermo-mechanical modeling toward a digital twin for thin-wall fused filament fabrication. In *Multifunctional Materials and Structures II*, volume 13947, pages 176–185. SPIE, 2026
- [7] Mumin Adhami, Mateo Garcia-Sandoval, Thienan Hoang, Matthew Whetham, Can Sun, Austin RJ Downey, Yanzhou Fu, and Lang Yuan. Design of vacuum-and pressure-compatible optical and thermal camera modules for laser powder bed fusion. In *Non-destructive Characterization and Monitoring of Advanced Materials, Aerospace, Civil Infrastructure, and Transportation XX*, volume 13950, pages 73–79. SPIE, 2026
- [8] Thienan Hoanga, Mumin Adhamia, Mateo E Garcia-Sandovala, Can Suna, and Austin RJ. In-situ thermal imaging for quality classification of dissimilar metal laser welds in battery tab welding. In *Thermosense: Thermal Infrared Applications XLVIII*, volume 14038, page 121. SPIE, 2026

Academic Reviewer

Journal Name	Times	Publisher
Additive Manufacturing	2	Elsevier
Journal of Manufacturing Processes	5	Elsevier
Advances in Manufacturing	2	Springer
IEEE Access	1	IEEE
Experimental Techniques	2	Wiley
Measurement	2	Elsevier
Scientia Iranica	1	Elsevier
Frontiers in Artificial Intelligence	1	Frontiers Media
International Journal of Advanced Manufacturing Technology	1	Springer
International Journal of Extreme Manufacturing	1	IOP Publishing
Mechanics Research Communications	1	Elsevier
Rapid Prototyping Journal	1	Emerald Publishing
3D Printing and Additive Manufacturing	2	Mary Ann Liebert
Industry 4.0 and Smart Manufacturing Conference	1	Elsevier

Student Mentor

Master's Students Currently Mentored

- [1] Mumin Adhami; Mechanical Engineering (USC), “Characterizing vibration patterns for defect detection in laser powder bed fusion”, 2024 - Present

Undergraduate Students Currently Mentored

- [1] Matthew Whetham; Mechanical Engineering (USC), “Unsupervised clustering analysis for real-time quality assurance in laser powder bed fusion”, 2024 - Present
- [2] Thienan Hoang; Mechanical Engineering (USC), “In situ monitoring for printer wearout”, 2024 - Present
- [3] Samuel Roberts; Mechanical Engineering (USC), “Modeling particle density with the discrete element method in laser powder bed fusion”, 2024 - Present
- [4] Dawodu, Teslim; Computer Engineering (BC), “Drone configuration for water quality sensing”, 2024 - Present
- [5] Guscott, Omario; Computer Engineering (BC), “Sensor mutual inspection for autonomous vehicle ”, 2024 - Present
- [6] Gauthier, Phumosny; Computer Engineering (BC), “Autonomous vehicle configuration”,

2024 - Present

- [7] Rolle, Charlton; Computer Engineering (BC), “Machine learning application in autonomous vehicle”, 2024 - Present

Undergraduate Students Previously Mentored

- [1] Stabler Charlie; Mechanical Engineering (USC), “Customized cover design for laser powder bed fusion”, 2019 - 2021
- [2] Ionita Sebastian; Mechanical Engineering (USC), “Thermal distribution in fused filament fabrication printing process”, 2019 - 2021
- [3] Flemming Malichi; Mechanical Engineering (USC), “In situ monitoring in laser powder bed fusion”, 2020 - 2021
- [4] Young Geoffrey; Mechanical Engineering (USC), “In-situ monitoring hardware configuration”, 2021 - 2022
- [5] Vaught Jacob; Computer Science and Engineering (USC), “Finite element analysis for additive manufacturing”, 2022 - 2023
- [6] Hung-Tien Huang; Computer Science and Engineering (USC), “Finite element model update and structural validation with machine learning”, 2019 - 2023
- [7] Priddy Braden; Mechanical Engineering (USC), “Real-time splatter tracking”, 2019 - 2023

Undergraduate Students Advised

- [1] Samuel Roberts; Aerospace Engineering, “Using the Discrete Element Method to Model Particle Density in Laser Powder Bed Fusion Dampers”, 2024 - 2025. Magellan Journey Scholar \$2,500. (Funded)
- [2] Charlie Buren; Mechanical Engineering, “Event Camera Paired In-situ Monitoring in Laser Powder Bed Fusion”, 2025 - 2026. McNair Fellow Scholar \$3,000. (Funded)
- [3] Matt Whetham; Mechanical Engineering, “Low-Cost In-Situ Monitoring and Unsupervised Clustering Analysis for Real-Time Quality Assurance in Laser Powder Bed Fusion”, 2024 - 2025. Magellan Mini Journey Scholar \$1,000. (Submitted, not awarded)

High School Students Mentored

- [1] Adelaide Rogers, “Particle damper packing simulation”, June - July 2023
- [2] Jaden Coffey, “Thermal distribution in battery charge process”, June - July 2023